Name of the Experiment..... Date .....

Page No. 1

Exp. Result ..... Exp. No. 1

Introduction

In the field of Engineering, it's necessary to obtain the unknown field parameters like displacement or forces, which can describe the behaviour of structures. These unknowns can be obtained by using exact analytical solns or energy methods, which are formulated with governing equilibrium eqns of analytical solutions eqns.

To reduce continuous system to a discrete one by conventional method, a lot of effort and assumption must be made, which are cumbersome and error prone.

1) Algebraic work corresponding to the solid mechanics is programmed in systematic matrix calculations and formulated into a mathematical model. User need not work with complicated algebra.

2) Ability to consider alternate loading and bending condn for the same geometry.

3) Accuracy of data mainly depends on the no. of approximations. Since in FEM we can make lesser no. of assumptions, degree of accuracy will be high with a min percentage of error.

FEM

Force method

Displacement method

Name of the Experiment..... Date .....

Exp. Result ..... Exp. No. .....

Page No.

a) **Force method:-** Here forces are unknown.

b) **Displacement Method:-** Here displacements are unknown.

1) **Preprocessing:-** It is the act of preparation of data such as nodal locations, element connectivity, co-ordinates, Imposing B.C, applications of load and providing material information to elements.

2) **Processing:-** This stage involves stiffness generation, strain, energy calculation and solns of eqns resulting in the evolution of nodal variable, induced elemental forces, strains and stresses.

3) **Post-processing:-** The geometric deformation and distribution of forces, stresses and strains in the structure, here the aspect of plotting a displaced plotting nodal data in the form of contour plots, such as isotherms and isobars and conversion of element oriented data into best fitting nodal values is discussed.

11/08/2025

Name of the Experiment..... Bar - 1 ..... Date ..... Page No. 3

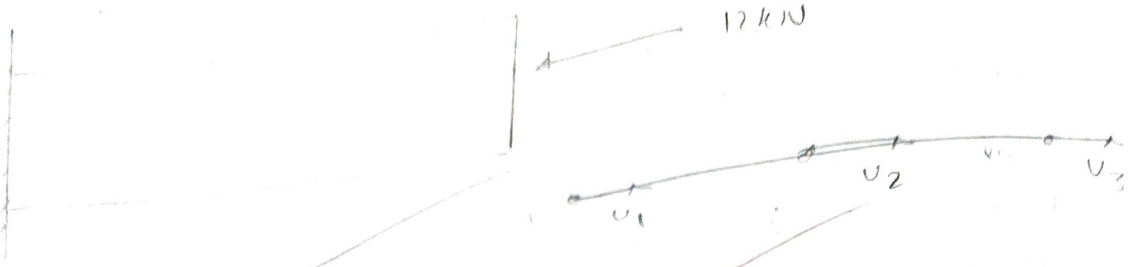
Exp. Result ..... Exp. No. ..... 1 .....

Expt No. 1

AIM:- Determine nodal displacements and axial stress in each element.

PROCEDURE:-

1) Utility menu > File > Change Jobname > Enter Bar Analysis.
2) Utility Menu > File > Change Title > Enter Bar Analysis.
3) Preprocessor > Element types > Add/Edit/Delete > Add > Link 1
4) Preprocessor > Real Constant > Add/Edit/Delete > Add > Ok > Area > 500
5) Preprocessor > Material Property > Young's Modulus > Enter E_x = 2e5.
6) Preprocessor > Modeling > Create > node > In active CS > Create three nodes at Location [0, 0, 0]; [500, 0, 0] and [1000, 0, 0].
7) Preprocessor > Load > Define Load > Apply > Structural > Displacement > Select Node no. 1 and apply 'ALL DOF'
8) Preprocessor > Load > Define Load > Apply > Structural > Force/moment on nodes > Select node number 3 and enter value f_x = -12000N.
9) Solution > Solve > Current LS
10) General Postprocessor > Plot results > Deform Shape.
11) Post processor > Contour Plot > Nodal solution > Displacement vector sum.
12) Postprocessor > Element Table > Define Table > Add > By sequence num > LS 1.
13) Postprocessor > Element Table > List Elem Table > LS 1 > Ok.



$$k^1 = \frac{AE}{L}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$= \frac{500 \times 200 \times 10^3}{0.500}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = 10^3\begin{bmatrix} 200 & -200 \\ -200 & 200 \end{bmatrix} = k^2$$

$$k = 10^3\begin{bmatrix} 200 & -200 & 0 \\ -200 & 400 & -200 \\ 0 & -200 & +200 \end{bmatrix}$$

$$F = \{\, 0 \quad 0 \quad -12000 \,\}^T$$

$$[k]\{\delta\} = \{f\}$$

$$10^3\begin{bmatrix} 200 & -200 & 0 \\ -200 & 400 & -200 \\ 0 & -200 & 200 \end{bmatrix}\begin{Bmatrix} U_1 \\ U_2 \\ U_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ -12000 \end{Bmatrix}$$

$$U_1 = 0$$

$$(400u_2 - 200u_3)\times 10^3 = 0$$

$$(-200u_2 + 200u_3)\times 10^3 = 12000$$

$u_2 = 0.06\,mm$, $u_3 = 0.12\,mm$

$$\sigma_1 = \frac{E}{L}\begin{bmatrix}-1 & 1\end{bmatrix}\begin{bmatrix}q_1\\ q_2\end{bmatrix} = \frac{200\times10^3}{500}\begin{bmatrix}-1 & 1\end{bmatrix}\begin{Bmatrix}0\\ 0.06\end{Bmatrix}$$

$$= 24\ MPa$$

$$\sigma_2 = \frac{E}{L}\begin{bmatrix}-1 & 1\end{bmatrix}\begin{bmatrix}q_2\\ q_3\end{bmatrix} = \frac{200\times10^3}{500}\begin{bmatrix}-1 & 1\end{bmatrix}\begin{Bmatrix}0.06\\ 0.12\end{Bmatrix}$$

$$= 400\left[-0.06 + 0.12\right]$$

$$= 24\ MPa$$

$$R = [k_1]\{q_1\} - \{F\}$$

$$= 200\times10^3 \times 0 - k_{11}u_1^{0} + k_{12}u_2 + k_{13}u_3^{0} - F_1$$

$$= 200\times10^3 \times 0.06 - 12000 - 0$$

$$= -12000\,N$$

Name of the Experiment..... Date .....

Exp. Result ..... Exp. No. .....

RESULT:-

	Ansys	Theoretical
Deformation	0.12	0.1177
Stress	24	23.89
Reaction	-12000	-1198.97

18/08/2015

Name of the Experiment..... Bar - 2 ..... Date .....

Page No. 5

Exp. Result ..... Exp. No. ..... 2 .....

Expt No. 2

AIM:- Determine the Nodal displacements and Reaction forces for the stepped Bar.

PROCEDURE:-

1) Utility Menu > File > Change Job Name
2) Utility menu > File > Change Title.
3) Preprocessor > ~~Real constant~~ Element Type > Add / Edit / Delete > Add > Link 1
4) Preprocessor > Real Constant > Add / Edit / Delete > Add > ok > Arra (Create 2 set of real constants.
5) Preprocessor > Material Property > Material Models > Young's Modulus > Enter, $E_x = 15e^6$ Under Material Models > Material > New Model > Enter $E_x = 10e6$
6) Preprocessor > Modeling > Create > Nodes > In active CS > Create 3 Nodes (0,0,0), (20,0,0) and (40,0,0).
7) Preprocessor > Modeling > Create > Element > Element Attributes
8) Preprocessor > Modeling > Create > Element > Auto Numbered > Thru Node.
9) Repeat procedure 7.
10) Preprocessor > Load > Define load > Apply > Structural > Displacement > on Nodes > Pick Node number 1 and apply 'All DOF.'
11) Preprocessor > load > Define Load > Apply > Structural > Force > on Nodes > Apply. $F_x = 20000$
12) Solution > Solve > Current LS.
13) General Postprocessor > Plot result > Deform shape.
14) Postprocessor > Contour plot > Nodal Solution > Displacement vector sum.
15) General Post processor > List result > Reaction > All structural force 'F'.



$$k^1 = \frac{AE}{l^e}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \frac{100 \times 15 \times 10^6}{20}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$= 10^6 \begin{bmatrix} 75 & -75 \\ -75 & 75 \end{bmatrix}$$

$$k^2 = \frac{60 \times 10 \times 10^6}{20}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = 10^6 \begin{bmatrix} 25 & -25 \\ -25 & 25 \end{bmatrix}$$

$$k = \begin{bmatrix} 75 & -75 & 0 \\ -75 & 100 & -25 \\ 0 & -25 & 25 \end{bmatrix}$$

$$F = \{ 0 \quad 0 \quad 20000 \}^T$$

$$Q = \{ U_1 \quad U_2 \quad U_3 \}^T$$

$$[k]\{Q\} = \{F\}$$

$$10^6 \begin{bmatrix} 75 & -75 & 0 \\ -75 & 100 & -25 \\ 0 & -25 & 25 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 20000 \end{Bmatrix}$$

$u_1 = 0$

$-75u_2 + 100u_3 \quad 100u_2 - 25u_3 = 0$

$$-25u_2 + 25u_3 = \frac{20000}{10^6}$$

$u_2 = 2.67 \times 10^{-4}$ mm, $\quad u_3 = 1.067 \times 10^{-3}$ mm

$R = \overset{0}{k_{11}u_1} + k_{12}u_2 + \overset{0}{k_{13}u_3} - F_1$

$= -75 \times 10^6 \times 2.67 \times 10^{-4}$

$= -20025\ N$

Name of the Experiment..... Date ..... Page No. 6

Exp. Result ..... Exp. No. .....

RESULT:-

$v_2 = 2.67 \times 10^{-4}$, $v_3 = 1.067 \times 10^{-3}$

$n = -2.0025\ N$

Seq

Cell

25/08/2015

Name of the Experiment..... Date ..... Page No. 7

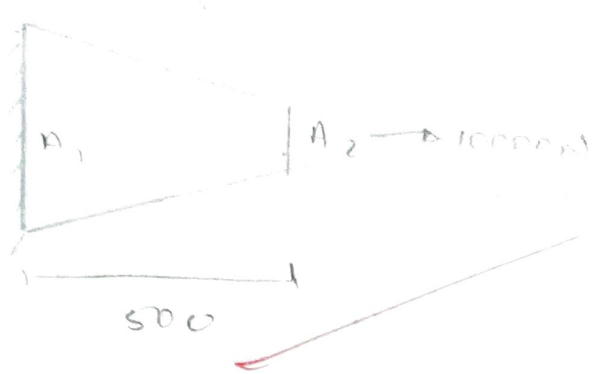
Exp. Result ..... Exp. No. .....

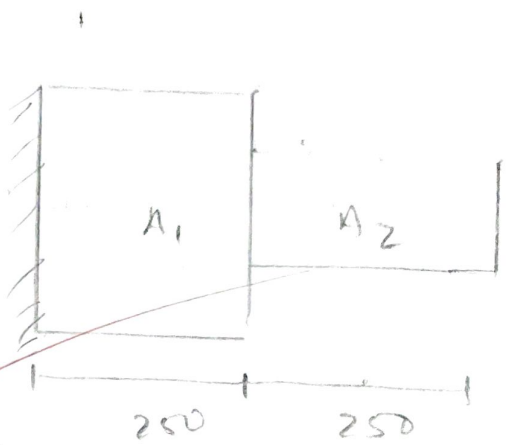
Experiment No. 3

AIM:- Obtain nodal displacements for the tapered bar.

PROCEDURE:-

1) Utility menu > File > Change Jobname.
2) Utility menu > File > Change Title
3) Preprocessor > Element Types > Add/Edit/Delete > Add > Link 1
4) Preprocessor > Real Constant > Add/Edit/Delete > Add > OK > Area > Create 2 set of Real Constant > $A_1 = 1000mm^2$, $A_2 = 500mm^2$.
5) Preprocessor > Material Property > Young's Modulus > Enter $E_x = 2 * 10e5$
6) Preprocessor > Modelling > Create nodes In active > Create 3 nodes > (0, 0, 0); (0, 0, 250); (0, 0, 500)
7) Preprocessor > Modelling > Create > Element > Elements Attributes.
8) Preprocessor > modelling > Create > Element > Auto numbered > Thru node > Create element blw nodes 1 and 2.
9) Repeat 7 and 8 for element 2.
10) Preprocessor > Load > Define > Apply > Structural > Displacement > Select node 1 and Apply 'All DOF' in the wizard.
11) Preprocessor > Load > Define Load > Apply > Structural > Force/Moments > On nodes > Select '3' apply force = 1000N.
12) Solution > Solve > Current Ls
13) Post processor > Contour Plot > Nodal solns > Displacement Vector Sum.





$A_1 = 1000$, $A_2 = 500$, $E = 2 \times 10^5$ MPa

$$k^1 = \frac{AE}{l_e}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \frac{1000 \times 2\times10^5}{250}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$= 10^3 \begin{bmatrix} 800 & -800 \\ -800 & 800 \end{bmatrix}$$

$$k^2 = \frac{500 \times 2\times10^5}{250}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = 10^3 \begin{bmatrix} 400 & -400 \\ -400 & 400 \end{bmatrix}$$

$$F = \{ 0 \quad 0 \quad 10000 \}^T$$

$$Q = \{ U_1 \quad U_2 \quad U_3 \}^T$$

$$K = \begin{bmatrix} 800 & -800 & 0 \\ -800 & 1200 & -400 \\ 0 & -400 & 400 \end{bmatrix}$$

$$[K]\{Q\} = \{F\}$$

$$\begin{bmatrix} 800 & -800 & 0 \\ -800 & 1200 & -400 \\ 0 & -400 & 400 \end{bmatrix} \begin{Bmatrix} U_1 \\ U_2 \\ U_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 10000 \end{Bmatrix}$$

$U_1 = 0$

$1200 v_2 - 400 v_3 = 0$

$-400 v_2 + 400 v_3 = 10000$

$v_2 = 12.5 \text{ mm}$, $v_3 = 37.5 \text{ mm}$

$R = \cancel{k_{11} v_1}^{0} + k_{12} v_2 + \cancel{k_{13} v_3}^{0} - f_1$

$= -800 \times 10^3 \times 12.5 - 0$

$= -10000 N$

Name of the Experiment..... Date ..... Page No. 9

Exp. Result ..... Exp. No. .....

RESULT:-

$v_2 = 17.5\,mm$, $v_3 = 37.5\,mm$

$R = -10000N$

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Name of the Experiment..... Date ..... Page No. 9

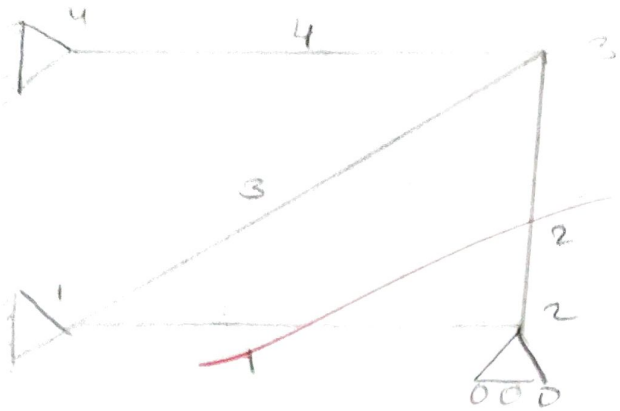
Exp. Result ..... Exp. No. 4

Experiment No. 4

AIM:- Calculate reaction forces, nodal displacement and element stresses for the truss.

PROCEDURE:-

1) Utility menu> File> Change Job Name
2) Utility menu> File> Change Title.
3) Preprocessor> Element Type> Add /Edit / Delete> Add Link 1.
4) Preprocessor> Real Constant> Add / Edit / Delete> Add> OK> Area 1.5
5) Preprocessor> Material property> Young's modulus> Enter $2 * 10e5$
6) Preprocessor> Modeling> Create> Node> In active CS> (reate 4 nodes 1 = (0,0,0); 2 (40,0,0); 3(40,30,0); 4(0,30,0).
7) Preprocessor> modelling> Create element> Auto numbered> Thro node> Create element b/w (1,3); (2,3); (3,4); (1,2)
8) Preprocessor> load > Define load> Apply> Structural > Displacement > On nodes > Select nodes 1 and 2.
9) Preprocessor> Load > Define Load> Apply> Structural > force/ Moments on nodes > F = 89 kN
10) Solution > Solve > Current LS
11) Post processor > Contour plot > Nodal solutions> displacement + vector Sum.
12) General postprocessor > List default > Reaction solutions> All stress force 'f'



Node no.	x	y
1	0	0
2	40	0
3	40	30
4	0	30

Element	Initial Node	Final Node
1	1	2
2	2	3
3	3	1
4	3	4

Length	Dirn	Cosine
40	1	0
30	0	1
50	-0.8	-0.6
40	-1	0

$$k^1 = 10^5 \begin{bmatrix} 5 & 0 & -5 & 0 \\ 0 & 0 & 0 & 0 \\ -5 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$k^2 = 10^5 \begin{bmatrix} 0 & 0 & -6.6 & 0 \\ 0 & 6.6 & 0 & -6.6 \\ 0 & 0 & +6.6 & 0 \\ 0 & -6.6 & 0 & 6.6 \end{bmatrix}$$

$$k^3 = 10^5 \begin{bmatrix} 5 & 0 & -5 & 0 \\ 0 & 0 & 0 & 0 \\ -5 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$k^4 = 10^5 \begin{bmatrix} 0.64 & 0.48 & -0.64 & -0.48 \\ 0.48 & -0.36 & -0.48 & 0.36 \\ -0.64 & -0.48 & 0.64 & 0.48 \\ -0.48 & 0.36 & 0.48 & -0.36 \end{bmatrix}$$

$$\sigma_1 = \frac{E}{l_e}\begin{bmatrix} -l & -m & l & m \end{bmatrix}\begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \frac{2\times10^5}{40}\begin{bmatrix} -1 & 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ 0.504 \\ 0.028 \end{bmatrix}$$

$$= 200\ \text{mPa}$$

$$\sigma_2 = \frac{E}{l_e} \quad \frac{2\times10^5}{30}\begin{bmatrix} -1 & 0 & 0 & 1 \end{bmatrix} = 0.2\ \text{mPa}$$

Name of the Experiment..... Date .....

Exp. Result ..... Exp. No. .....

RESULT:-

σ_1 = 200 mPa

σ_2 = 0.2 mPa

$R_1 = 0$ $R_4 = 75$ kN

$R_2 = 105$ kN $R_5 = R_6 = 0$

$R_3 = 75$ kN

15/09/2025

Name of the Experiment..... Date ..... Page No. 11

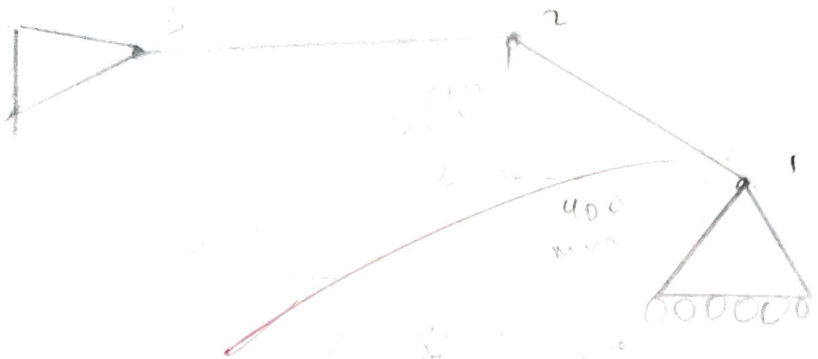
Exp. Result ..... Exp. No. .....

Experiment No. 5

Aim:- For given truss, determine the reaction forces, nodal displacements and element stresses

PROCEDURE:-

1) Utility menu > File > Change Job name
2) Utility menu > File > Change Title.
3) Preprocessor > Element Type > Add/Edit/Delete > Add > Link 1.
4) Preprocessor > Real Constant > Add/Edit/Delete > Add > OK > Area 200.
5) Preprocessor > Materials property > Young's modulus > Enter 70 * 10e6
6) Preprocessor > Modelling > Create > Node > In active CS > Create 3 nodes, Node 1(0,0,0) ; 2(500,0,0) ; 3(900, 300, 0)
7) Preprocessor > Modelling > Create > Element > Auto numbered > Thru node > Create element blw nodes (1,2) and (2,3)
8) Preprocessor > Load > Define load > Apply > Structural > Displacement > On nodes > Select nodes 1
9) Preprocessor > Load > Define load > Apply > Structural > Force/moments on nodes > Select node 2 and apply a force F_y = 12000N
10) Soln > Solve > Current LS
11) General postprocessor > Plot > Deform shape.
12) Post processor > Contour Plot > Nodal soln > Displacement vector sum
13) General postprocessor > List result > Reaction soln > All struc force 'F'.



1) Node No.

Node No.	X	Y
1	0	0
2	-400	300
3	900	300
4	0	0

2)

Element	I Node	F Node	Length	Dirn (l)	Cosines (m)
1	1	2	500	0.8	0.6
2	2	3	500	1	0
3	3	1	50	-0.8	-0.6
4	4	4	40	-1	0

$$K = \frac{AE}{L}\begin{bmatrix} l^2 & lm & -l^2 & -lm \\ ml & m^2 & -ml & -m^2 \\ -l^2 & -lm & l^2 & lm \\ -ml & -m^2 & ml & m^2 \end{bmatrix}$$

$$l = \frac{x_f - x_i}{L_e} = \frac{400 - 0}{500} = 0.8 \quad , \quad m = \frac{y_f - y_i}{L_e} = \frac{300 - 0}{500} = 0.6$$

$$l = \frac{x_f - x_i}{L_e} = \frac{900 - 400}{500} = 1 \quad , \quad m = 0$$

$$k^1 = \frac{180 \times 10^{-4} \times 200 \times 10^3}{0.5} \begin{bmatrix} 0.64 & 0.48 & -0.64 & -0.48 \\ 0.48 & 0.36 & -0.48 & -0.36 \\ -0.64 & -0.48 & 0.64 & 0.48 \\ -0.48 & -0.36 & 0.48 & 0.36 \end{bmatrix}$$

$$= \begin{bmatrix} 38400 & 28800 & -38400 & -28800 \\ 28800 & 21600 & -28800 & -21600 \\ -38400 & -28800 & 38400 & 28800 \\ -28800 & -21600 & 28800 & 21600 \end{bmatrix}$$

$$k^2 = \begin{bmatrix} 60000 & 0 & -60000 & 0 \\ 0 & 0 & 0 & 0 \\ -60000 & 0 & +60000 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \times 4$$

$$k = \begin{bmatrix} 38400 & 28800 & -38400 & -28800 & 0 & 0 \\ 28800 & 21600 & -28800 & -21600 & 0 & 0 \\ -38400 & -28800 & 98400 & 288000 & -60000 & 0 \\ -28800 & -21600 & 28800 & 21600 & 0 & 0 \\ 0 & 0 & -60000 & 0 & 60000 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\{k\}\{\delta\} = \{F\}$$

$$\begin{bmatrix} 38400 & 28800 & -38400 & -28800 & 0 & 0 \\ 28800 & 21600 & -28800 & -21600 & 0 & 0 \\ -38400 & -28800 & 98400 & 28800 & & \\ -28800 & -21600 & 28800 & 21600 & & \\ 0 & 0 & -60000 & 60000 & & \\ 0 & 0 & 0 & 0 & & \end{bmatrix} \begin{Bmatrix} v_1 \\ v_1 \\ v_2 \\ v_2 \\ v_3 \\ v_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 12000 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$$

$v_2 = v_2 = v_3 = v_3 = 0$

$$38400 u_1 + 29800 v_1 = 0$$
$$29800 u_1 + 21600 v_1 = 0$$

$$u_1 = 0.04$$
$$v_1 = -0.13$$

$$\sigma_1 = \frac{2 \times 10^5}{800} \begin{bmatrix} -1 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0.04 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= 10 \text{ N/mm}^2$$

$$\sigma_2 = 0$$

$$R = [k]\{\delta\} - \{F\}$$

$$R_2 = -16000 \text{ N}$$

$$R_3 = 13979.2$$

$$R_6 = R_4 = R_5 = 0$$

Name of the Experiment..... Date ..... Page No. 12

Exp. Result ..... Exp. No. .....

RESULT:-

$q_3 = 0.04$

$q_4 = -0.13$

$\sigma_1 = 10\ N/mm^2$

$\sigma_2 = -75\ N/mm^2$

$R_1 = -1600\ N$

$R_3 = +3997.2\ N$

06/10/2025

Name of the Experiment..... Date ..... Page No. 14

Exp. Result ..... Exp. No. .....

Experiment No. 6

Aim:- Find the deflection and slope of the beam in the figure.

PROCEDURE:-

1) Utility menu > File > Change Job Name
2) Utility Menu > File > Change Title
3) Preprocessor > Element type > Add/Edit/Delete > Add > Beam 3
4) Preprocessor > Real constants > Add/Edit/Delete > Add > OK > Area = 100 > Area Moment of Inertia = 833.33
5) Preprocessor > Material property > material models > Young's modulus > Enter E_x = 2e5
6) Preprocessor > material Property > Create > Node > In active CS > Create three nodes 1(0,0,0); 2(1000,0,0); 3(2000,0,0)
7) Preprocessor > Modelling > Create > Element > Auto numbered > Thru node > Create element.
8) Preprocessor > Load > Define load > Apply > Structural > Displacement > On nodes > Select node 1 > Apply ALL DOFs
9) Preprocessor > Load > Define load > Apply > Structural > Pressure > On beam > select
10) Solution > Solve > Current LS
11) General postprocessor > Plot results > Deform shape*
12) Postprocessor > Contour plots > Nodal solution > Displacement Vector Sum
13) To find Bending moment
14) General postprocessor > Contour Plot > Line element resolution.



$E = 70\ GPa, \quad I = 8333.33\ mm^4$

$$k^1 = \frac{70 \times 10^3 \times 833.33}{1^3} \begin{bmatrix} 12 & +6 & -12 & 6 \\ 6 & 4 & -6 & 2 \\ -12 & -6 & 12 & -6 \\ 6 & 2 & -6 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 700 & 350 & -700 & 350 \\ 350 & 233.33 & -350 & 116.67 \\ -700 & -350 & 700 & -350 \\ 350 & 116.67 & -350 & 233.33 \end{bmatrix} \begin{matrix} 1,3 \\ 2,4 \\ 3,5 \\ 4,6 \end{matrix} = k^2$$

$$k = \begin{bmatrix} 700 & 350 & -700 & 350 & 0 & 0 \\ 350 & 233.33 & -350 & 116.67 & 0 & 0 \\ -700 & -350 & 1400 & 0 & -700 & 350 \\ 350 & 116.67 & 0 & 233.33 & -350 & 116.67 \\ 0 & 0 & -700 & -350 & 700 & -350 \\ 0 & 0 & 350 & 116.67 & -350 & 233.33 \end{bmatrix}$$

$$F^1 = 10^3 \begin{bmatrix} -1 & -0.333 & -1 & 0.333 \end{bmatrix}^T$$

$$F^2 = 10^3 \begin{bmatrix} -2 & -3 & -4.5 & 0.2 \end{bmatrix}^T$$

$$\begin{bmatrix} 700 & 350 & -700 & 350 & 0 & 0 \\ 350 & 233.33 & -350 & 116.67 & 0 & 0 \\ -700 & -350 & 1400 & 0 & -700 & 350 \\ 350 & 116.67 & 0 & 233.33 & -350 & 116.67 \\ 0 & 0 & -700 & 350 & 700 & -350 \\ 0 & 0 & 350 & 116.67 & -350 & 233.33 \end{bmatrix} \begin{Bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \\ v_3 \\ \theta_3 \end{Bmatrix} = \begin{Bmatrix} -10^4 \\ -333 \\ 2.080 \\ 33.33 \\ -458 \\ 200 \end{Bmatrix}$$

$v_1 = \theta_1 = v_2 = \theta_2 = v_3 = 0$

$233.33\,\theta_3 = 200$

$\therefore \theta_3 = 0.86$ radians

$v_{max} = 4_3\,\theta_3$

$= \frac{1}{4}(-1 - 2 + 3 + 1) \times \theta_3$

$= \frac{1}{4} \times 0.86 = 0.0214$ m

Name of the Experiment..... Date ..... Page No. 15

Exp. Result ..... Exp. No. .....

RESULT:-

θ_3 = 0.86 radians

V_{max} = 0.0214 m/s

13/10/2025

Name of the Experiment..... Date ..... Page No. 16

Exp. Result ..... Exp. No. .....

Experiment No. 7

Determine the temp distribution through the composite wall and heat loss to ambient temp.

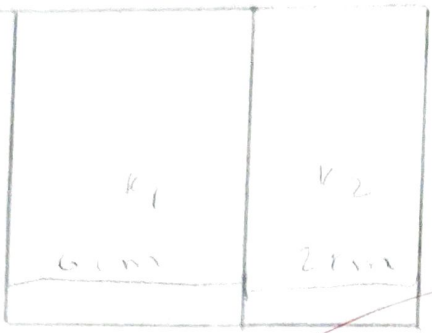
PROCEDURE:-

1) Utility menu > File > Change Job Name
2) Utility Menu > File > Change Title.
3) Preprocessor > material property > Add/Edit/Delete > Add > Thermal > Solid > Planes
4) Preprocessor > material Property > Thermal > Conductivity > Isotropic > Enter K_1 = 6 W/mK
5) Preprocessor > Modelling > Create > Area > Rectangle > By dimension > Give dimension, $x_2 = 10$, $y_2 = 10$
6) Preprocessor > ~~Modelling~~ > Load > Apply > Thermal > Temp
7) Solutions > Solve > Current L.S
8) General Postprocessor > Plot result > Contour plot > Nodal Solutions > DOF soln > Nodal Temp

RESULT:-

	Ansys	Theory
~~T_1~~ T_2	0.833°C	0.845°C
T_3	- 2.9167°C	- 2.918°C

27/10/2025



$T_1 = 20^\circ C$, $k_1 = 6$ w/mk , $k_2 = 20$ w/mk

$h = 1000$ w/mk , $T_\infty = -5^\circ C$

$$k^1 = \frac{kA}{\ell}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \frac{6 \times 1}{0.060}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 100 & -100 \\ -100 & 100 \end{bmatrix}$$

$$k^2 = \frac{20 \times 1}{0.02}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1000 & -1000 \\ -1000 & 1000 \end{bmatrix}$$

$$k = \begin{bmatrix} 100 & -100 & 0 \\ -100 & 1100 & -1000 \\ 0 & -1000 & 1000 \end{bmatrix}$$

$$F = hAT_\infty \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ -5000 \end{Bmatrix}$$

$$[k]\{T\} = \{F\}$$

$$\begin{bmatrix} 100 & -100 & 0 \\ -100 & 1100 & -1000 \\ 0 & -1000 & 1000 \end{bmatrix} \begin{Bmatrix} T_1 \\ T_2 \\ T_3 \end{Bmatrix} \begin{Bmatrix} 0 \\ 0 \\ -5000 \end{Bmatrix}$$

$$-100T_1 + 1100T_2 - 1000T_3 = 0$$

$$1100T_2 - 1000T_3 = 100T_1$$

$$1100T_2 - 1000T_3 = 2000.$$

$$1100T_2 - 1000T_3 = 2000$$

$$-1000T_2 + 1000T_3 = -5000$$

$$T_2 = 0.833^\circ C$$

$$T_3 = -2.9167^\circ C$$

Name of the Experiment..... Date ..... Page No. 18

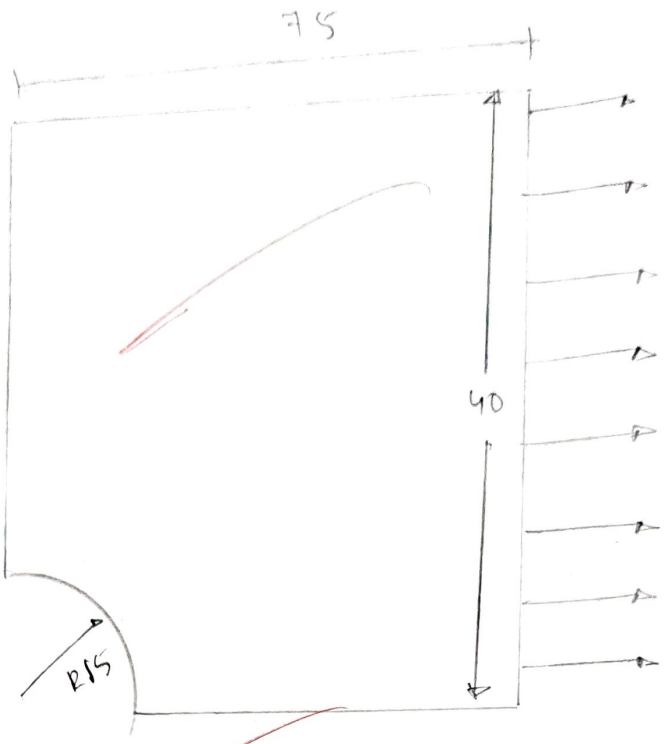
Exp. Result ..... Exp. No. .....

Experiment No. 8

AIM:- Determine the location and magnitude of the max stress in the plate.

PROCEDURE:-

1) Utility menu > file > Change Job Name
2) Utility menu > file > Change Title.
3) Preprocessor > Element types > Add/Edit/Delete > Add > solid > Quad 4 node and 2.
4) Preprocessor > Material property > Young's modulus > Enter E = 2e5 and poisson's ratio = 0.3
5) The model is symmetric we are going to create 1/4 of the model.
6) Preprocessor > modelling > Create > Area > Circle > Solid Circle > Define dimensions > Radius = 15, X = 75, Y = 40
7) Preprocessor > meshing > mesh tool > Smart size > ON > free > Quad > mesh
8) Preprocessor > Load > Apply > Structural > Displacement > Symmetry B.C > Select line on x coordinates zero and y coordinates zero and apply symmetry.
9) Preprocessor > Load > Apply > Structural > Pressure > On line > Enter > ALL DOF.
10) Solution > Solve > Current LS
11) General postprocessor > Plot result > Deform shape.
12) General postprocessor > Plot result > Contour plate > Nodal soln > Displacement Vector sum.
13) General post processor > Plot result > Contour plot > Nodal solution > Stress > Von Mises Stress



$p = 100 N/mm^2$

E5 $E = 30 C 6$

$p_{m_x} = 0.046 N/mm$

$p_{m_y} = 359.64 N/mm^2$

$p_{m_v} = 13.57 N/mm^2$

Name of the Experiment..... Date ..... Page No. 19

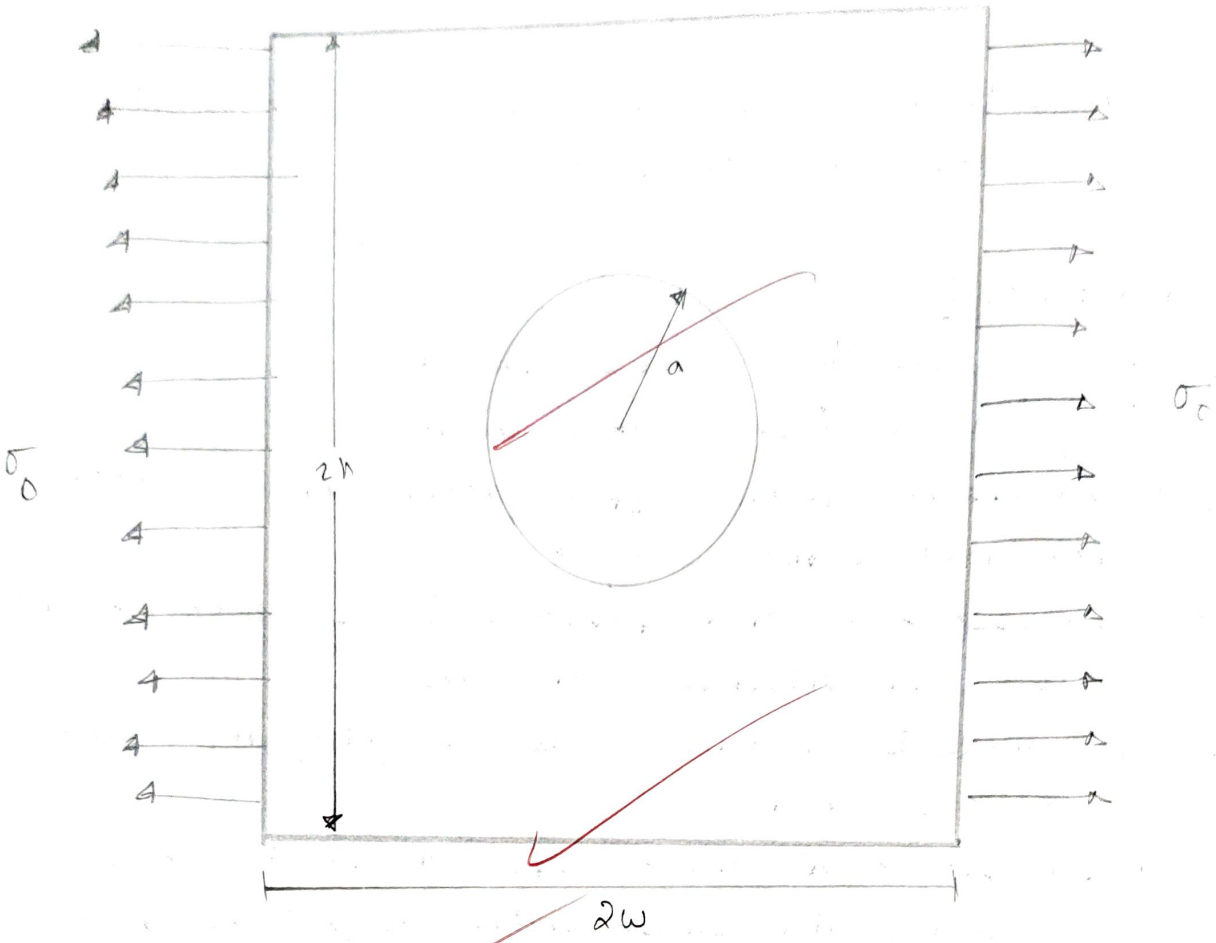
Exp. Result ..... Exp. No. .....

Experiment No. 9

Aim:- Stress Analysis of rectangular plate with hole, Let $2w = 100$ mm, $2h = 250$ mm, $a = 10$ mm, $\sigma_0 = 100$ MPa

PROCEDURE:-

1) Utility Menu > Change To File > Change Job Name
2) Utility Menu > File > Change Title.
3) Preprocessor > Element type > Add/Edit/Delete > Add > Solid > Quad and node 1, 2.
4) Preprocessor > material property > Young's Modulus > Enter E = 2e5 and poisson's ratio = 0.3.
5) The model is symmetric we are going to create 1/4 of the model.
6) Preprocessor > Modeling > Operate > Booleans > Subtract > Area > Select the bigger area first circle' apply and select smaller area and click OK.
7) Preprocessor > Meshing > mesh tool > Smart size > ON > Lines > Quad > mesh > Pick all
8) Preprocessor > Load > Apply > structural > Displacement > Symmetry B.C > Select x coordinates 0 and y coordinates 0 and apply symmetric B.C's.
9) Preprocessor > Load > Apply > Structural > Pressure > On Line > Enter value = 100
10) Solution > Solve > Current L.S. 10/11/2025
11) General post processor > Plot result > Deform Shape.
12) General post processor > Plot result > Contour plot > Nodal Soln > Displacement Vector Sum.
13) General postprocessor > plot result > Contour plot > Nodal soln. > Stress > Von mises stress.



Thickness = 10 mm

D_{max} = 0.04303 N/m

S_{mY} = 323.85 N/mm²

S_{mN} = 14.66 N/mm²

Name of the Experiment..... Date ..... Page No. 20

Exp. Result ..... Exp. No. .....

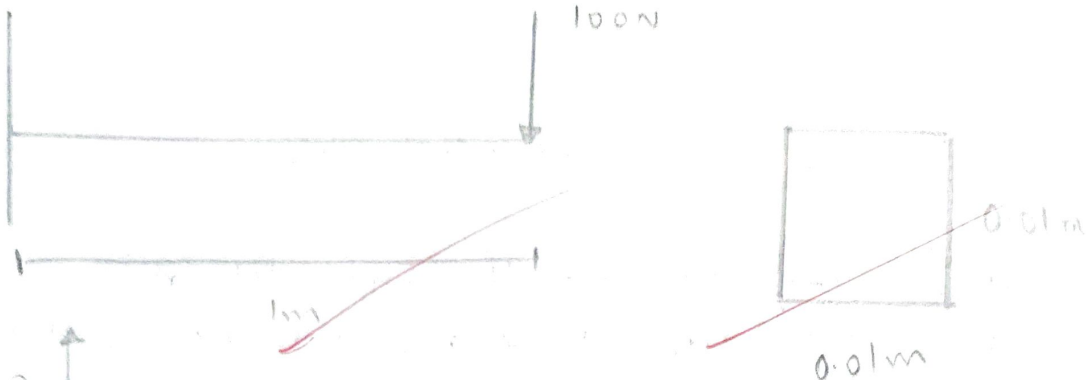
Experiment No.10

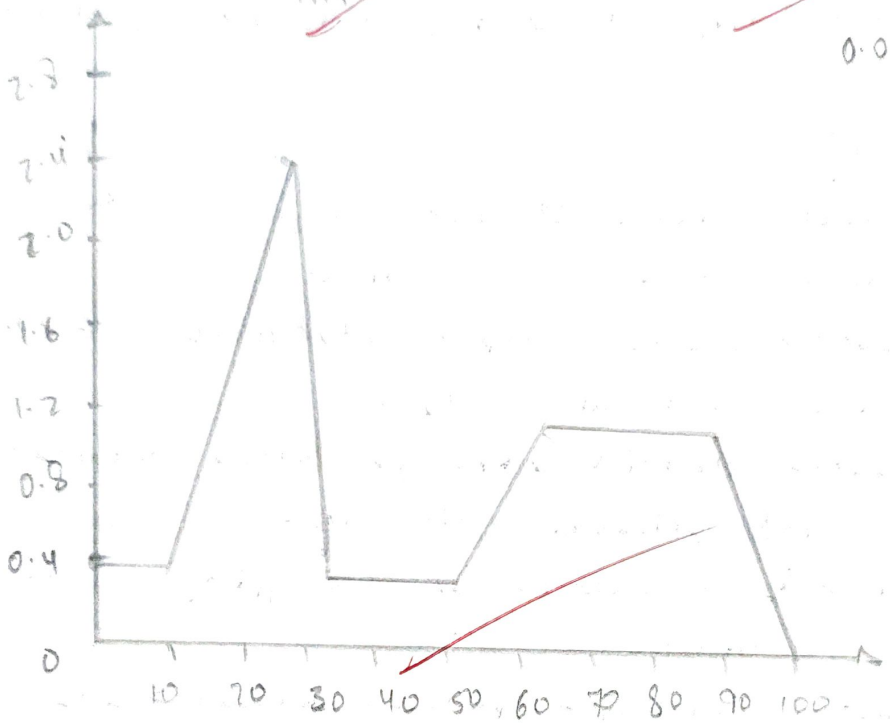
AIM:- To determine SFM of a cantilever beam

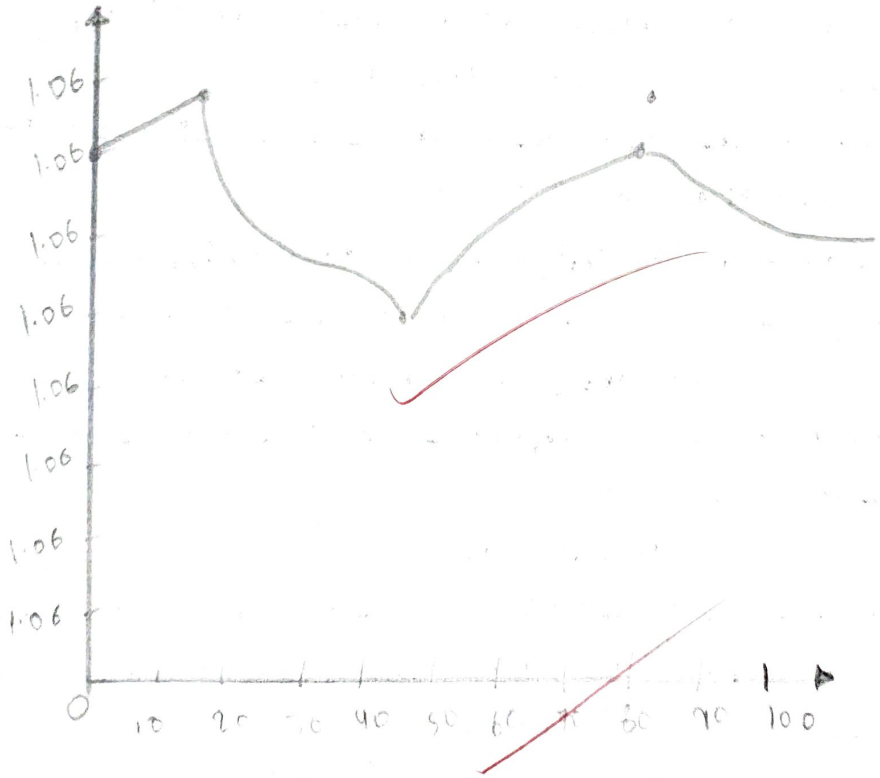
PROCEDURE:-

1) Preference > structural
2) Preprocessor > Element types > Add > Beam material property > Material model > Structural > Linear > Isotropic E = 206500 e6 > Density = 7830
3) Sections > Beams > Common sections > b = 0.01 and h = 0.01
4) modelling > Create func > straight line > Meshing > Analysis types > New analysis > Harmonic > Loads > displays
5) Load > Load steps > Faq > ~~sub~~ stepped out > solve current L.S
6) Post processor > Add > History constants > Dof solns > displacement > Graph data > plot control > Style > Graph > Model area > x-axis > logarithmic.

17/11/2025


100N
1m
0.01m
0.01m


2.4
2.0
1.6
1.2
0.8
0.4
0
10
20
30
40
50
60
70
80
90
100


1.06
1.06
1.06
1.06
1.06
1.06
1.06
1.06
0
10
20
30
40
50
60
70
80
90
100